Moon-Drop Observation-Mode Calibration

Flash, marks, and a playful Bayesian forensic analysis

Eric Novik

2026-08-25

Table of contents

1 Aim

Students were instructed to record the flashes produced as a falling object crossed fixed distance marks. A second interface condition exposed the marks directly. This report uses labeled instructor trials to calibrate the timing signature of the two observation modes and then asks which mode each student experiment most resembles.

The analysis is intentionally playful. The quantity being classified is the *timing pattern produced by the interface*, not anyone's moral character.

The workflow is:

1. Read and audit every observational file.
2. Use labeled instructor runs to compare flash and marks timing behavior.
3. Fit a hierarchical calibration model in Stan.
4. Fit every student trial in Stan.
5. Compare each student's effective intercept draws with the calibrated mode distributions.
6. Inspect rankings and every fitted experiment visually.

2 Data assembly

2.1 Labeled instructor calibration trials

Instructor filenames encode both the known observation mode and the true gravitational condition. For example, `gravity-flash-124-01.csv` is a flash trial generated with $g = 1.24$ m/s^2.

```
calibration_files <- list.files(
  data_dir,
  pattern = "^gravity-(flash|marks)-[0-9]+-[0-9]+[.]csv$",
  full.names = TRUE
)

calibration <- map_dfr(calibration_files, function(file) {
  metadata <- str_match(
    basename(file),
    "^gravity-(flash|marks)-([0-9]+)-([0-9]+)[.]csv$"
  )

  mode <- metadata[, 2]
  g_true <- as.numeric(metadata[, 3]) / 100
  experiment <- str_remove(basename(file), "[.]csv$")

  read_csv(file, show_col_types = FALSE) |>
    transmute(
      experiment,
      mode,
      g_true,
      trial,
      mark,
      distance_m,
      observed_time_s,
      sqrt_distance = sqrt(distance_m),
      physical_time_s = sqrt(2 * distance_m / g_true),
      apparent_delay_s = observed_time_s - physical_time_s
```

```
    )
})

calibration_map <- calibration |>
  distinct(experiment, mode, g_true) |>
  arrange(mode, experiment) |>
  mutate(
    calibration_trial = row_number(),
    mode_id = match(mode, mode_levels),
    beta_true = sqrt(2 / g_true)
  )

calibration <- calibration |>
  left_join(
    calibration_map,
    by = c("experiment", "mode", "g_true")
  )

calibration_inventory <- calibration |>
  group_by(experiment, mode, g_true) |>
  summarise(
    observations = n(),
    first_mark = min(mark),
    last_mark = max(mark),
    .groups = "drop"
  )

knitr::kable(
  calibration_inventory,
  digits = 2,
  booktabs = TRUE,
  caption = "Labeled instructor calibration trials"
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 1: Labeled instructor calibration trials

experiment	mode	g_true	observations	first_mark	last_mark
gravity-flash-124-01	flash	1.24	17	1	20
gravity-flash-124-02	flash	1.24	16	1	20
gravity-flash-162-01	flash	1.62	17	1	20
gravity-marks-132-01	marks	1.32	18	2	20
gravity-marks-143-01	marks	1.43	18	2	19

2.2 Student files

All student CSVs share the columns `trial`, `mark`, `distance_m`, and `observed_time_s`. Rosie Bai originally submitted a PNG, which was transcribed into `Rosie Bai-gravity-unknown-observations.csv`, and later supplied a different CSV, retained unchanged as `Rosie Bai-gravity-unknown-observations-2.csv`. Because each source labels its sole experiment `trial = 1`, the ingestion step maps the later CSV to Rosie's trial 2 while leaving both source files unchanged.

```
student_files <- list.files(
  data_dir,
  pattern = "^[A-Z].*[.]csv$",
  full.names = TRUE
)

students <- map_dfr(student_files, function(file) {
  file_name <- basename(file)
  student <- str_remove(file_name, "-gravity.*$") |>
    stringi::stri_trans_nfc()
  source_format <- if_else(
    file_name == "Rosie Bai-gravity-unknown-observations.csv",
    "PNG transcription",
    "CSV"
  )

  read_csv(file, show_col_types = FALSE) |>
    transmute(
      student,
      source_file = file_name,
      source_format,
      trial = if (
        file_name == "Rosie Bai-gravity-unknown-observations-2.csv"
      ) 2L else as.integer(trial),
      mark,
      distance_m,
      observed_time_s,
      sqrt_distance = sqrt(distance_m)
    )
})

student_trial_map <- students |>
  distinct(student, trial, source_file, source_format) |>
  arrange(student, trial) |>
  mutate(student_trial = row_number())

students <- students |>
  left_join(
    student_trial_map,
    by = c("student", "trial", "source_file", "source_format")
  )

student_inventory <- students |>
  group_by(student, source_format) |>
  summarise(
    trials = n_distinct(trial),
```

```r
    observations = n(),
    first_distance_m = min(distance_m),
    last_distance_m = max(distance_m),
    .groups = "drop"
  )

knitr::kable(
  student_inventory,
  booktabs = TRUE,
  longtable = TRUE,
  caption = "Student experiment inventory"
) |>
  kable_styling(latex_options = c("repeat_header"))
```

Table 2: Student experiment inventory

student	source_format	trials	observations	first_distance_m	last_distance_m
Alice Zhang	CSV	1	18	5	95
Annie Kiyonaga	CSV	3	52	5	100
Ben Bozza	CSV	1	20	5	100
Elliot Weinbaum Suárez	CSV	6	102	5	100
Emily Marshall	CSV	1	19	5	100
Haocheng Zhang	CSV	1	19	5	100
Haydy Rodriguez Lopez	CSV	1	19	5	100
Jihyun Park	CSV	3	50	5	100
Kaya Sittinger	CSV	1	20	5	100
Melody Fang	CSV	1	19	5	100
Noa Kipnis	CSV	5	87	5	100
Raffae Mansuri	CSV	1	20	5	100
Rosie Bai	CSV	1	15	5	100
Rosie Bai	PNG transcription	1	10	30	100
Victoria Maleci	CSV	3	57	5	100

The assembled data contain 86 labeled calibration observations from 5 instructor trials and 527 observations from 30 student trials contributed by 14 students.

3 Physics and observation models

For distance x, true physical elapsed time is

$$\tau(x; g) = \sqrt{\frac{2x}{g}}.$$

The stopwatch records an observation process rather than physical time directly. Its linearized representation is

$$t_i^{\mathrm{obs}} = \alpha_j + \beta_j \sqrt{x_i} + \varepsilon_i.$$

Here α_j is an **effective intercept signature**. It is useful for separating the interfaces, but it need not be a literal constant human reaction delay: intercept and slope can trade off when the delay changes during a drop.

For labeled calibration trial j, true gravity is known, so

$$\beta_j^{\mathrm{true}} = \sqrt{\frac{2}{g_j}}.$$

We can therefore inspect the raw apparent delay directly:

$$d_{ij} = t_{ij}^{\mathrm{obs}} - \tau(x_{ij}; g_j).$$

3.1 Apparent delay over the drop

```r
calibration |>
  ggplot(
    aes(
      physical_time_s,
      apparent_delay_s,
      color = mode,
      group = experiment
    )
  ) +
  geom_hline(yintercept = 0, linewidth = 0.25) +
  geom_line(alpha = 0.55, linewidth = 0.5) +
  geom_point(size = 1) +
  geom_smooth(
    aes(group = mode),
    method = "lm",
    se = FALSE,
    linewidth = 1
  ) +
  scale_color_manual(values = mode_colors) +
  labs(
    x = "true physical elapsed time (s)",
    y = "observed minus physical time (s)",
    color = NULL,
    title = "Flash delay falls during the drop; marks delay is comparatively flat"
  )
```

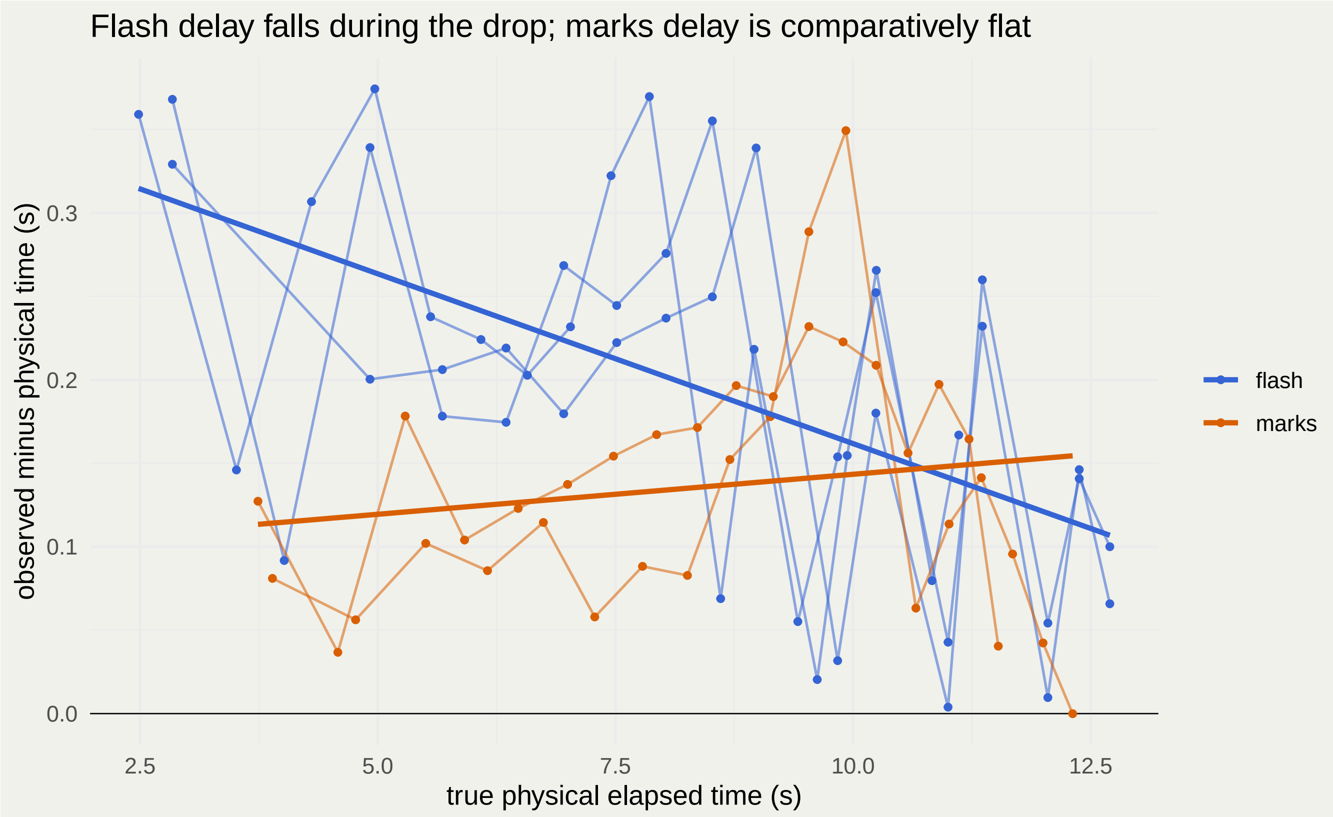


Figure 1: Apparent delay relative to the known physical curve.

```
calibration_descriptive <- calibration |>
  group_by(experiment, mode, g_true) |>
  group_modify(function(.x, .g) {
    delay_fit <- lm(apparent_delay_s ~ physical_time_s, data = .x)
    observation_fit <- lm(observed_time_s ~ sqrt_distance, data = .x)

    tibble(
      effective_intercept_s = unname(coef(observation_fit)[1]),
      observed_slope = unname(coef(observation_fit)[2]),
      apparent_delay_trend = unname(coef(delay_fit)[2]),
      residual_sd_s = sigma(observation_fit)
    )
  }) |>
  ungroup()

knitr::kable(
  calibration_descriptive,
  digits = 3,
  booktabs = TRUE,
  caption = "Descriptive calibration fits before Stan"
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 3: Descriptive calibration fits before Stan

experiment	mode	g_true	effective_intercept_s	observed_slope	apparent_delay_trend	residua
gravity-flash-124-01	flash	1.24	0.358	1.245	-0.020	
gravity-flash-124-02	flash	1.24	0.353	1.245	-0.019	
gravity-flash-162-01	flash	1.62	0.377	1.087	-0.021	
gravity-marks-132-01	marks	1.32	0.090	1.235	0.003	
gravity-marks-143-01	marks	1.43	0.084	1.193	0.009	

```r
calibration_descriptive |>
  ggplot(aes(mode, effective_intercept_s, color = mode)) +
  geom_point(size = 3, position = position_jitter(width = 0.04)) +
  scale_color_manual(values = mode_colors, guide = "none") +
  labs(
    x = NULL,
    y = "effective intercept (s)",
    title = "Observed calibration signature"
  )
```

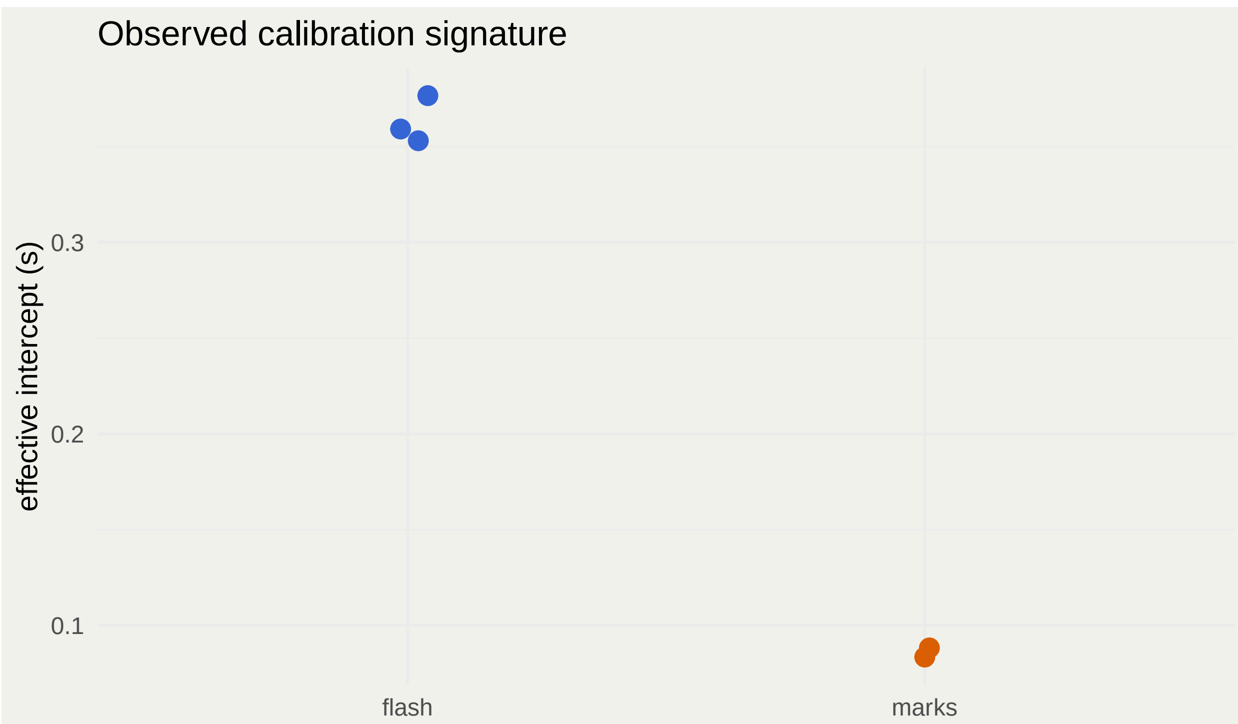


Figure 2: The free-slope effective intercept cleanly separates the labeled instructor trials.

4 Hierarchical Stan calibration

The calibration model allows trial-specific intercepts but pools them by known mode:

$$\alpha_j \sim \mathrm{Normal}(\mu_{\alpha,m_j}, \tau_\alpha).$$

A shared τ_α is used because only two marks trials and three flash trials are available. This prevents one mode from winning classifications merely by acquiring a much wider, poorly estimated tail.

The trial slope is centered on the true physical slope plus a mode-specific slope bias:

$$\beta_j = \beta_j^{\text{true}} + \mu_{\Delta\beta,m_j} + u_{\beta,j}.$$

The observation likelihood is Gaussian:

$$t_{ij}^{\text{obs}} \sim \text{Normal}\left(\alpha_j + \beta_j\sqrt{x_{ij}}, \sigma_{m_j}\right).$$

```
calibration_data <- list(
  N = nrow(calibration),
  J = nrow(calibration_map),
  M = length(mode_levels),
  trial = calibration$calibration_trial,
  mode = calibration_map$mode_id,
  z = calibration$sqrt_distance,
  y = calibration$observed_time_s,
  beta_true = calibration_map$beta_true
)
```

```
calibration_model <- cmdstan_model(
  file.path(stan_dir, "delay-calibration.stan")
)

calibration_fit <- calibration_model$sample(
  data = calibration_data,
  seed = 20260825,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  init = 0,
  refresh = 0,
  adapt_delta = 0.995
)
```

4.1 Calibration draws with `tidybayes`

```
calibration_mode_draws <- calibration_fit |>
  spread_draws(
    mu_alpha[mode_id],
    mu_beta_bias[mode_id],
    sigma[mode_id],
    tau_alpha,
    tau_beta_bias
```

```r
  ) |>
  mutate(mode = mode_levels[mode_id])

calibration_trial_draws <- calibration_fit |>
  spread_draws(
    alpha[calibration_trial],
    beta[calibration_trial],
    g_effective[calibration_trial]
  ) |>
  left_join(calibration_map, by = "calibration_trial")

calibration_summary <- calibration_mode_draws |>
  ungroup() |>
  select(
    .draw,
    mode,
    mu_alpha,
    mu_beta_bias,
    sigma,
    tau_alpha
  ) |>
  pivot_longer(
    cols = c(mu_alpha, mu_beta_bias, sigma, tau_alpha),
    names_to = "parameter",
    values_to = "value"
  ) |>
  group_by(mode, parameter) |>
  median_qi(value, .width = 0.90) |>
  transmute(
    mode,
    parameter,
    median = value,
    lower = .lower,
    upper = .upper
  )

knitr::kable(
  calibration_summary,
  digits = 3,
  booktabs = TRUE,
  caption = "Posterior calibration summaries (90 percent intervals)"
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 4: Posterior calibration summaries (90 percent intervals)

mode	parameter	median	lower	upper
flash	mu_alpha	0.355	0.282	0.429
flash	mu_beta_bias	-0.023	-0.035	-0.012

flash	sigma	0.091	0.077	0.110
flash	tau_alpha	0.021	0.002	0.072
marks	mu_alpha	0.091	0.010	0.173
marks	mu_beta_bias	0.006	-0.006	0.019
marks	sigma	0.073	0.060	0.091
marks	tau_alpha	0.021	0.002	0.072

```r
calibration_mode_draws |>
  ggplot(aes(mu_alpha, mode, fill = mode)) +
  stat_halfeye(.width = c(0.50, 0.90), alpha = 0.75) +
  scale_fill_manual(values = mode_colors, guide = "none") +
  labs(
    x = expression(mu[alpha] ~ "(seconds)"),
    y = NULL,
    title = "Calibrated effective-intercept centers"
  )
```

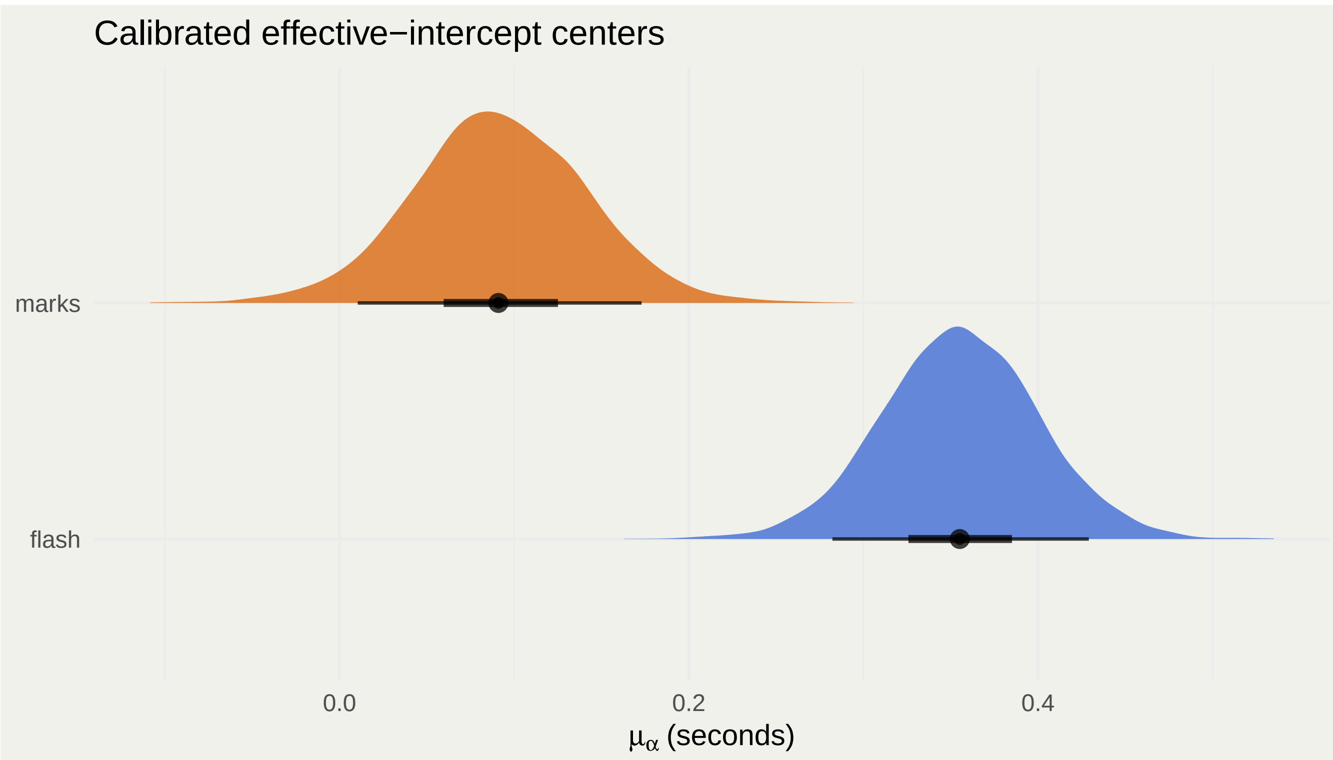


Figure 3: Posterior mode centers for the effective intercept.

The posterior median mode centers are approximately 0.091 seconds for marks and 0.355 seconds for flashes.

```r
calibration_predictive <- calibration_mode_draws |>
  ungroup() |>
  mutate(alpha_new = rnorm(n(), mu_alpha, tau_alpha))
```

```r
calibration_predictive |>
  ggplot(aes(alpha_new, fill = mode, color = mode)) +
  geom_density(alpha = 0.25, linewidth = 0.8) +
  scale_fill_manual(values = mode_colors) +
  scale_color_manual(values = mode_colors) +
  labs(
    x = "effective intercept for a new trial (s)",
    y = "posterior predictive density",
    fill = NULL,
    color = NULL,
    title = "Calibration distributions used for the game"
  )
```

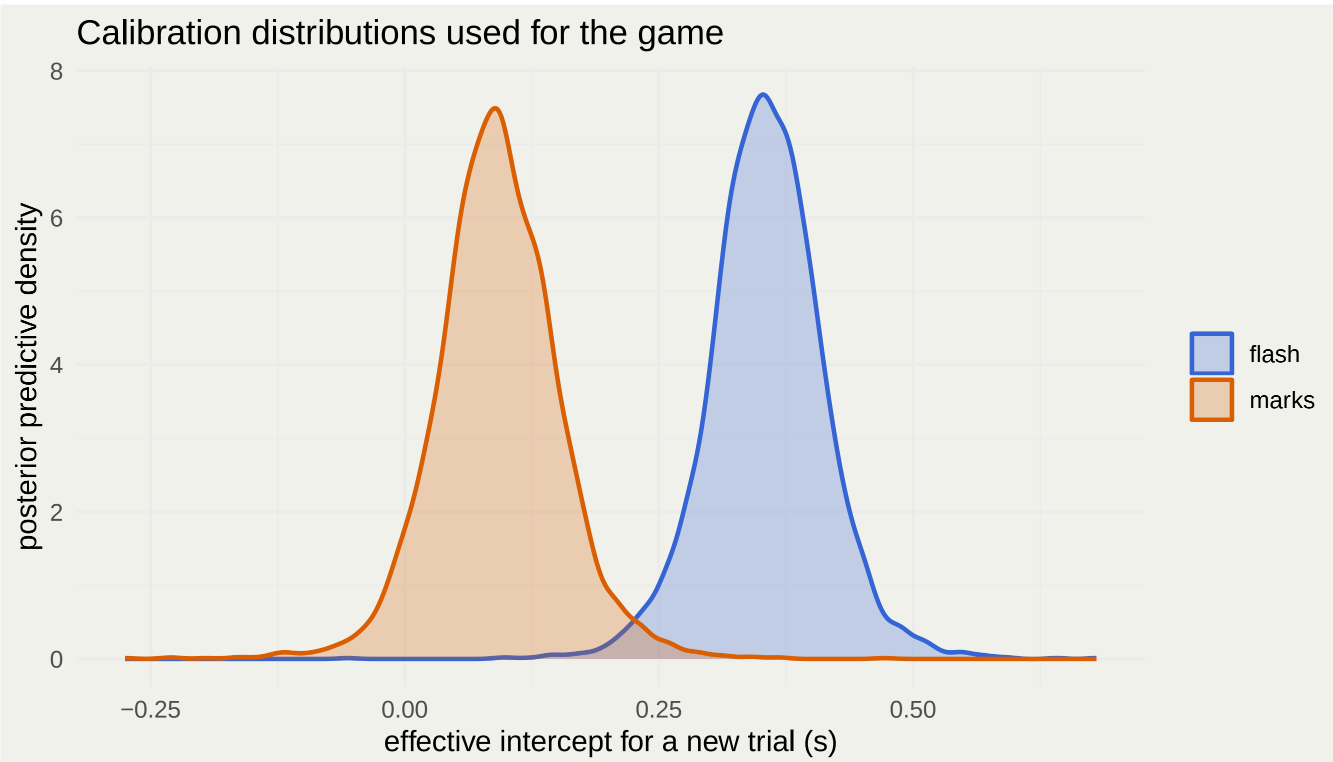


Figure 4: Posterior predictive distributions for the effective intercept of a new trial.

4.2 Sampler diagnostics

```r
calibration_diagnostics <- calibration_fit$diagnostic_summary()

calibration_parameter_diagnostics <- calibration_fit$summary(
  variables = c(
    "mu_alpha",
    "tau_alpha",
    "mu_beta_bias",
    "tau_beta_bias",
```

```
    "sigma"
  )
) |>
  as_tibble() |>
  transmute(
    variable,
    rhat,
    ess_bulk,
    ess_tail
  )

knitr::kable(
  calibration_parameter_diagnostics,
  digits = 1,
  booktabs = TRUE,
  caption = "Calibration-model convergence diagnostics"
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 5: Calibration-model convergence diagnostics

variable	rhat	ess_bulk	ess_tail
mu_alpha[1]	1	2999.4	2765.7
mu_alpha[2]	1	2582.9	2217.1
tau_alpha	1	1720.6	1777.3
mu_beta_bias[1]	1	2577.8	2293.1
mu_beta_bias[2]	1	2285.8	1850.0
tau_beta_bias	1	1337.3	1609.9
sigma[1]	1	3580.8	2291.0
sigma[2]	1	4354.0	2382.9

The calibration fit produced 0 divergent transitions and 0 maximum-treedepth transitions.

5 Student trial model

Each student trial receives its own intercept, slope, and noise scale:

$$t_{ij}^{\text{obs}} \sim \text{Normal}\left(\alpha_j + \beta_j \sqrt{x_{ij}}, \sigma_j\right), \qquad \hat{g}_j = \frac{2}{\beta_j^2}.$$

The trials are fitted jointly for computational convenience, but the priors do not pool one student's regression coefficients toward another's.

```r
student_data <- list(
  N = nrow(students),
  J = nrow(student_trial_map),
  trial = students$student_trial,
  z = students$sqrt_distance,
  y = students$observed_time_s
)
```

```r
student_model <- cmdstan_model(
  file.path(stan_dir, "student-trials.stan")
)

student_fit <- student_model$sample(
  data = student_data,
  seed = 20260826,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  init = 0,
  refresh = 0,
  adapt_delta = 0.95
)
```

```r
student_draws <- student_fit |>
  spread_draws(
    alpha[student_trial],
    beta[student_trial],
    sigma[student_trial],
    g_hat[student_trial]
  ) |>
  left_join(student_trial_map, by = "student_trial")

student_trial_summary <- student_draws |>
  ungroup() |>
  select(
    .draw,
    student_trial,
    student,
    trial,
    alpha,
    beta,
    g_hat,
    sigma
  ) |>
  pivot_longer(
    cols = c(alpha, beta, g_hat, sigma),
    names_to = "parameter",
    values_to = "value"
  ) |>
```

```r
  group_by(student_trial, student, trial, parameter) |>
  median_qi(value, .width = 0.90) |>
  transmute(
    student,
    trial,
    parameter,
    median = value,
    lower = .lower,
    upper = .upper
  )
```

5.1 Student-model diagnostics

```r
student_diagnostics <- student_fit$diagnostic_summary()

student_parameter_diagnostics <- student_fit$summary(
  variables = c("alpha", "beta", "sigma")
) |>
  as_tibble() |>
  summarise(
    maximum_rhat = max(rhat),
    minimum_bulk_ess = min(ess_bulk),
    minimum_tail_ess = min(ess_tail)
  )

knitr::kable(
  student_parameter_diagnostics,
  digits = 1,
  booktabs = TRUE,
  caption = "Student-model convergence summary"
) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 6: Student-model convergence summary

maximum_rhat	minimum_bulk_ess	minimum_tail_ess
1	3078.3	2100.7

The student fit produced 0 divergent transitions and 0 maximum-treedepth transitions.

6 Flash-versus-marks score

Stan does not sample a discrete observation-mode indicator. Instead, the mode is marginalized after fitting by comparing each effective-intercept draw with the calibrated predictive distributions.

For mode m and student trial j, the draw-specific log predictive score is

$$\ell_{m,j}^{(d)} = \log \mathrm{Normal}\left(\alpha_j^{(d)} \mid \mu_{\alpha,m}^{(d)}, \tau_\alpha^{(d)}\right).$$

For students with multiple trials, scores are added under the playful assumption that the same observation strategy was used across their trials. Equal prior odds are assigned to flash and marks:

$$P(\text{marks} \mid \text{data}) = \text{logit}^{-1}\left(\ell_{\text{marks}} - \ell_{\text{flash}}\right).$$

6.1 Draw-level classification

```
calibration_wide <- calibration_mode_draws |>
  ungroup() |>
  select(.draw, mode, mu_alpha, tau_alpha) |>
  pivot_wider(
    names_from = mode,
    values_from = mu_alpha,
    names_glue = "mu_alpha_{mode}"
  )

trial_classification_draws <- student_draws |>
  left_join(calibration_wide, by = ".draw") |>
  mutate(
    log_marks = dnorm(
      alpha,
      mu_alpha_marks,
      tau_alpha,
      log = TRUE
    ),
    log_flash = dnorm(
      alpha,
      mu_alpha_flash,
      tau_alpha,
      log = TRUE
    ),
    p_marks = plogis(log_marks - log_flash)
  )

student_classification_draws <- trial_classification_draws |>
  group_by(.draw, student) |>
  summarise(
    mean_alpha = mean(alpha),
    log_marks = sum(log_marks),
    log_flash = sum(log_flash),
    .groups = "drop"
  ) |>
  mutate(
```

```
    p_marks = plogis(log_marks - log_flash)
  )
```

6.2 Trial-level results

```
trial_results <- trial_classification_draws |>
  group_by(student_trial, student, trial) |>
  summarise(
    delay_median = median(alpha),
    delay_lower = quantile(alpha, 0.05),
    delay_upper = quantile(alpha, 0.95),
    g_median = median(g_hat),
    g_lower = quantile(g_hat, 0.05),
    g_upper = quantile(g_hat, 0.95),
    p_marks = mean(p_marks),
    .groups = "drop"
  ) |>
  arrange(desc(p_marks))

trial_results_display <- trial_results |>
  transmute(
    Student = student,
    Trial = trial,
    `Effective delay, s` = sprintf(
      "%.3f [%.3f, %.3f]",
      delay_median,
      delay_lower,
      delay_upper
    ),
    `Estimated g` = sprintf(
      "%.3f [%.3f, %.3f]",
      g_median,
      g_lower,
      g_upper
    ),
    `P(marks-like)` = scales::percent(p_marks, accuracy = 0.1)
  )

knitr::kable(
  trial_results_display,
  booktabs = TRUE,
  longtable = TRUE,
  caption = "Trial-level posterior summaries and marks-like scores"
) |>
  kable_styling(
    latex_options = c("repeat_header"),
    font_size = 8
  )
```

Table 7: Trial-level posterior summaries and marks-like scores

Student	Trial	Effective delay, s	Estimated g	P(marks-like)
Alice Zhang	1	0.116 [0.054, 0.180]	1.597 [1.573, 1.623]	97.6%
Jihyun Park	3	0.064 [-0.083, 0.213]	1.410 [1.363, 1.459]	94.7%
Jihyun Park	2	0.090 [-0.035, 0.226]	1.413 [1.372, 1.461]	92.9%
Rosie Bai	1	0.261 [-0.200, 0.715]	1.646 [1.489, 1.834]	44.9%
Raffae Mansuri	1	0.301 [0.204, 0.399]	1.344 [1.315, 1.374]	13.5%
Elliot Weinbaum Suárez	3	0.338 [0.221, 0.461]	1.670 [1.619, 1.722]	7.6%
Victoria Maleci	3	0.375 [0.226, 0.517]	1.292 [1.252, 1.334]	5.9%
Annie Kiyonaga	2	0.391 [0.236, 0.543]	1.377 [1.331, 1.425]	4.5%
Emily Marshall	1	0.390 [0.247, 0.532]	1.685 [1.625, 1.748]	4.4%
Annie Kiyonaga	3	0.356 [0.250, 0.459]	1.365 [1.333, 1.399]	3.6%
Victoria Maleci	1	0.378 [0.253, 0.500]	1.285 [1.249, 1.322]	3.5%
Jihyun Park	1	0.457 [0.267, 0.641]	1.508 [1.443, 1.578]	2.9%
Rosie Bai	2	0.397 [0.266, 0.526]	1.696 [1.639, 1.758]	2.7%
Annie Kiyonaga	1	0.508 [0.321, 0.684]	1.397 [1.337, 1.457]	1.3%
Noa Kipnis	4	0.471 [0.318, 0.616]	1.390 [1.342, 1.439]	1.1%
Haocheng Zhang	1	0.643 [0.404, 0.874]	1.843 [1.733, 1.961]	0.6%
Elliot Weinbaum Suárez	4	0.455 [0.332, 0.574]	1.733 [1.679, 1.785]	0.5%
Victoria Maleci	2	0.476 [0.354, 0.593]	1.331 [1.295, 1.368]	0.4%
Noa Kipnis	5	0.547 [0.403, 0.694]	1.409 [1.361, 1.459]	0.3%
Ben Bozza	1	0.437 [0.374, 0.496]	1.493 [1.471, 1.515]	0.2%
Elliot Weinbaum Suárez	5	0.501 [0.376, 0.624]	1.727 [1.673, 1.783]	0.2%
Elliot Weinbaum Suárez	1	0.582 [0.440, 0.727]	1.782 [1.714, 1.857]	0.1%
Kaya Sittinger	1	0.506 [0.409, 0.597]	1.402 [1.371, 1.433]	0.1%
Melody Fang	1	0.495 [0.393, 0.601]	1.408 [1.375, 1.443]	0.1%
Elliot Weinbaum Suárez	6	0.532 [0.415, 0.643]	1.747 [1.694, 1.799]	0.1%
Haydy Rodriguez Lopez	1	0.555 [0.448, 0.661]	1.967 [1.911, 2.023]	0.1%
Elliot Weinbaum Suárez	2	0.584 [0.466, 0.705]	1.778 [1.723, 1.835]	0.1%
Noa Kipnis	3	0.612 [0.505, 0.720]	1.438 [1.403, 1.475]	0.1%
Noa Kipnis	2	0.636 [0.511, 0.759]	1.453 [1.411, 1.496]	0.1%
Noa Kipnis	1	0.669 [0.525, 0.813]	1.447 [1.398, 1.499]	0.0%

6.3 Student-level ranking

```
student_ranking <- student_classification_draws |>
  group_by(student) |>
  summarise(
    delay_median = median(mean_alpha),
    delay_lower = quantile(mean_alpha, 0.05),
    delay_upper = quantile(mean_alpha, 0.95),
    p_marks = mean(p_marks),
    .groups = "drop"
  ) |>
  mutate(
    result = case_when(
      p_marks >= 0.80 ~ "marks-like",
      p_marks <= 0.20 ~ "flash-like",
      TRUE ~ "mixed / ambiguous"
    )
  ) |>
  arrange(desc(p_marks))
```

```
ranking_display <- student_ranking |>
  transmute(
    Rank = row_number(),
    Student = student,
    `Effective delay, s` = sprintf(
      "%.3f [%.3f, %.3f]",
      delay_median,
      delay_lower,
      delay_upper
    ),
    `P(marks-like)` = scales::percent(p_marks, accuracy = 0.1),
    Result = result
  )

knitr::kable(
  ranking_display,
  booktabs = TRUE,
  longtable = TRUE,
  caption = "Student-level marks-likeness ranking"
) |>
  kable_styling(
    latex_options = c("repeat_header"),
    font_size = 8
  )
```

Table 8: Student-level marks-likeness ranking

Rank	Student	Effective delay, s	P(marks-like)	Result
1	Alice Zhang	0.116 [0.054, 0.180]	97.6%	marks-like
2	Jihyun Park	0.204 [0.115, 0.295]	62.3%	mixed / ambiguous
3	Rosie Bai	0.327 [0.093, 0.564]	23.3%	mixed / ambiguous
4	Raffae Mansuri	0.301 [0.204, 0.399]	13.5%	flash-like
5	Emily Marshall	0.390 [0.247, 0.532]	4.4%	flash-like
6	Haocheng Zhang	0.643 [0.404, 0.874]	0.6%	flash-like
7	Annie Kiyonaga	0.417 [0.328, 0.505]	0.2%	flash-like
8	Ben Bozza	0.437 [0.374, 0.496]	0.2%	flash-like
9	Kaya Sittinger	0.506 [0.409, 0.597]	0.1%	flash-like
10	Melody Fang	0.495 [0.393, 0.601]	0.1%	flash-like
11	Haydy Rodriguez Lopez	0.555 [0.448, 0.661]	0.1%	flash-like
12	Victoria Maleci	0.410 [0.332, 0.483]	0.1%	flash-like
13	Elliot Weinbaum Suárez	0.499 [0.448, 0.549]	0.0%	flash-like
14	Noa Kipnis	0.586 [0.526, 0.644]	0.0%	flash-like

The most marks-like student-level timing signature is **Alice Zhang**, with posterior score 97.6%. Multiple-trial students can have mixed trial-level patterns, so the detailed table and diagnostic plots are more informative than the single ranking alone.

```
calibration_centers <- calibration_mode_draws |>
  group_by(mode) |>
  summarise(center = median(mu_alpha), .groups = "drop")

marks_center <- calibration_centers |>
```

```r
  filter(mode == "marks") |>
  pull(center)
flash_center <- calibration_centers |>
  filter(mode == "flash") |>
  pull(center)

student_ranking |>
  mutate(student = fct_reorder(student, delay_median)) |>
  ggplot(aes(delay_median, student, color = p_marks)) +
  geom_vline(
    xintercept = marks_center,
    color = mode_colors[["marks"]],
    linewidth = 0.7,
    linetype = "dashed"
  ) +
  geom_vline(
    xintercept = flash_center,
    color = mode_colors[["flash"]],
    linewidth = 0.7,
    linetype = "dashed"
  ) +
  geom_errorbar(
    aes(xmin = delay_lower, xmax = delay_upper),
    orientation = "y",
    width = 0,
    color = "grey45"
  ) +
  geom_point(size = 2.5) +
  scale_color_gradient(
    low = mode_colors[["flash"]],
    high = mode_colors[["marks"]],
    limits = c(0, 1),
    labels = scales::percent
  ) +
  labs(
    x = "student mean effective intercept (s), 90% interval",
    y = NULL,
    color = "P(marks-like)",
    title = "Student timing signatures"
  )
```

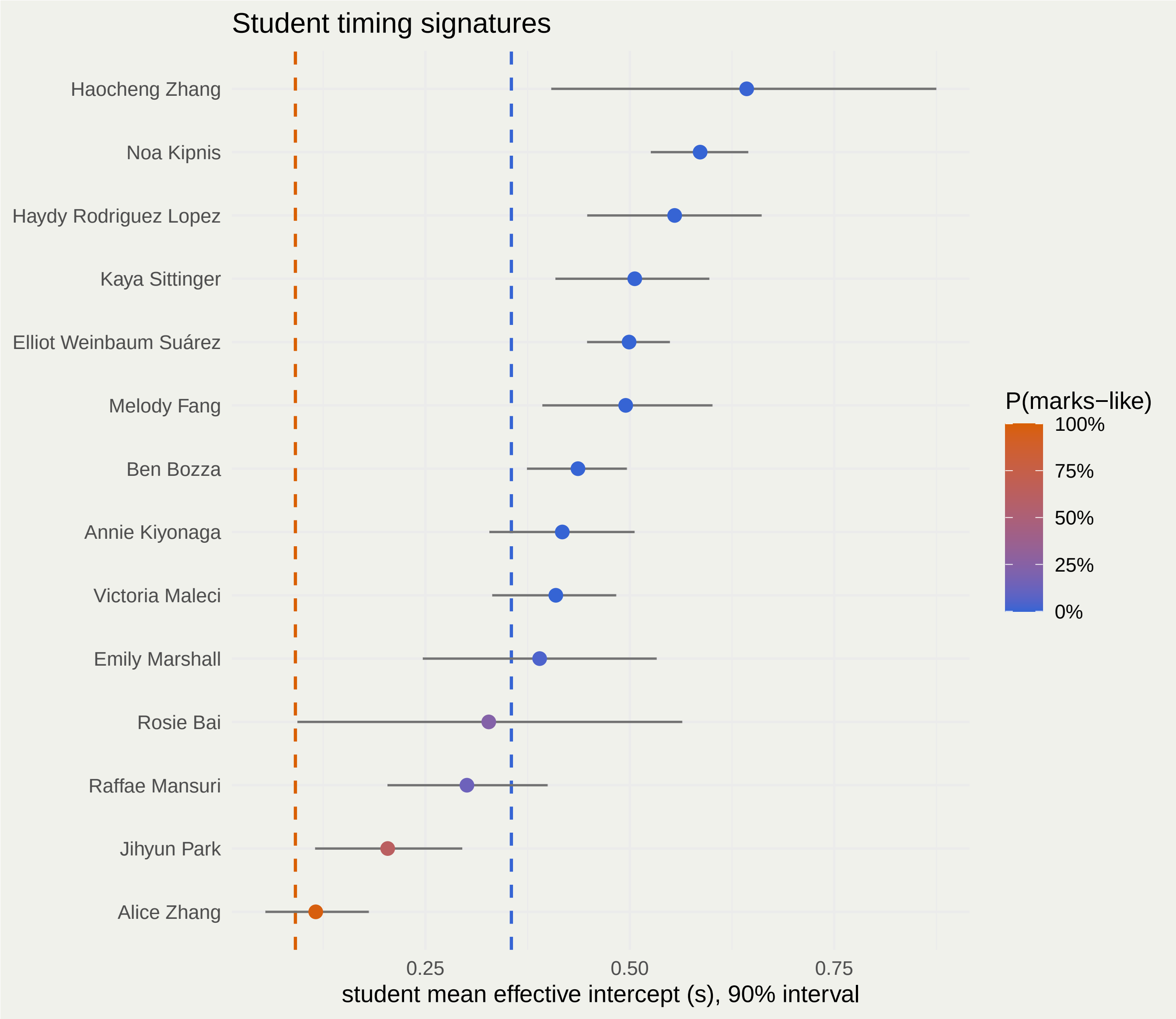


Figure 5: Student effective-delay estimates compared with calibrated mode centers.

```
student_ranking |>
  mutate(student = fct_reorder(student, p_marks)) |>
  ggplot(aes(p_marks, student, fill = result)) +
  geom_col(width = 0.7) +
  geom_vline(xintercept = c(0.20, 0.80), linetype = "dashed") +
  scale_x_continuous(
    limits = c(0, 1),
    labels = scales::percent
  ) +
  scale_fill_manual(
    values = c(
      "flash-like" = mode_colors[["flash"]],
      "mixed / ambiguous" = "grey60",
      "marks-like" = mode_colors[["marks"]]
    )
```

```
  ) +
  labs(
    x = "posterior P(marks-like)",
    y = NULL,
    fill = NULL,
    title = "The playful leaderboard"
  ) +
  theme(legend.position = "bottom")
```

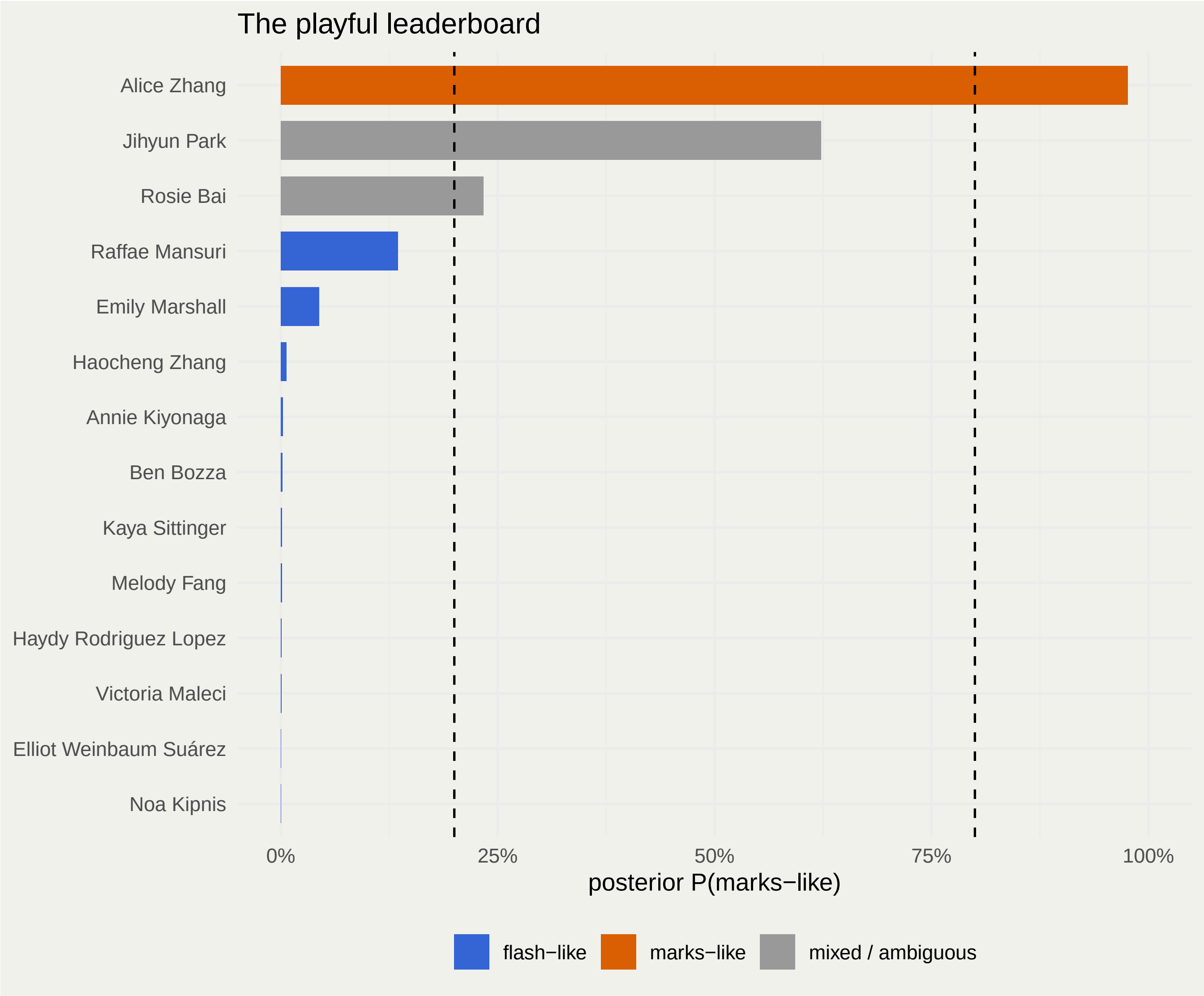


Figure 6: Posterior marks-like score under equal prior odds.

```
trial_results |>
  mutate(
    trial_label = paste0("trial ", trial),
    student = fct_reorder(student, p_marks, .fun = max)
  ) |>
  ggplot(aes(trial_label, student, fill = p_marks)) +
```

```r
  geom_tile(color = "white", linewidth = 0.5) +
  geom_text(
    aes(label = scales::percent(p_marks, accuracy = 1)),
    size = 2.6
  ) +
  scale_fill_gradient(
    low = mode_colors[["flash"]],
    high = mode_colors[["marks"]],
    limits = c(0, 1),
    labels = scales::percent
  ) +
  labs(
    x = NULL,
    y = NULL,
    fill = "P(marks-like)",
    title = "Trial-by-trial observation-mode signature"
  )
```

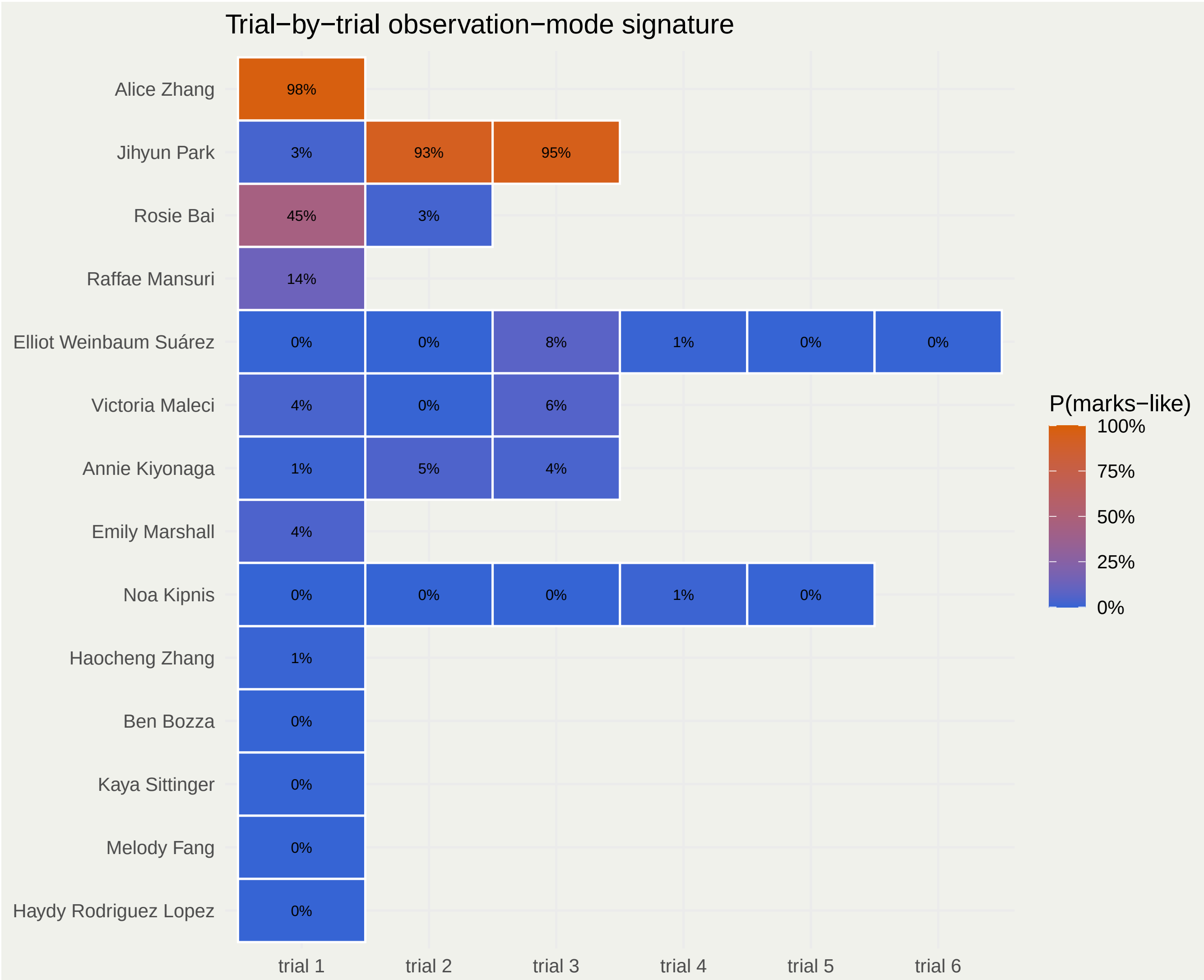


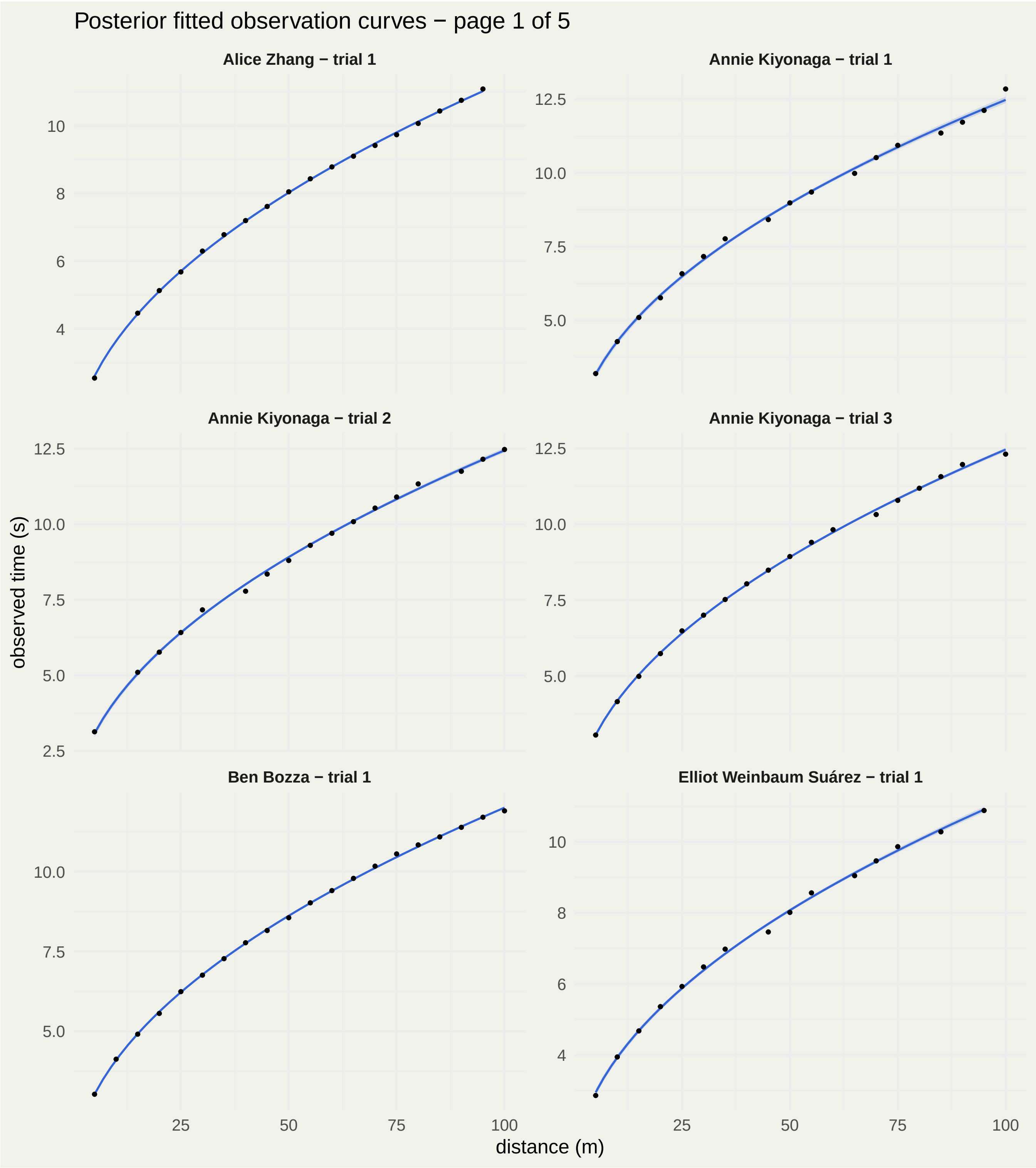
Figure 7: Trial-level mode scores reveal mixed strategies within some multi-trial submissions.

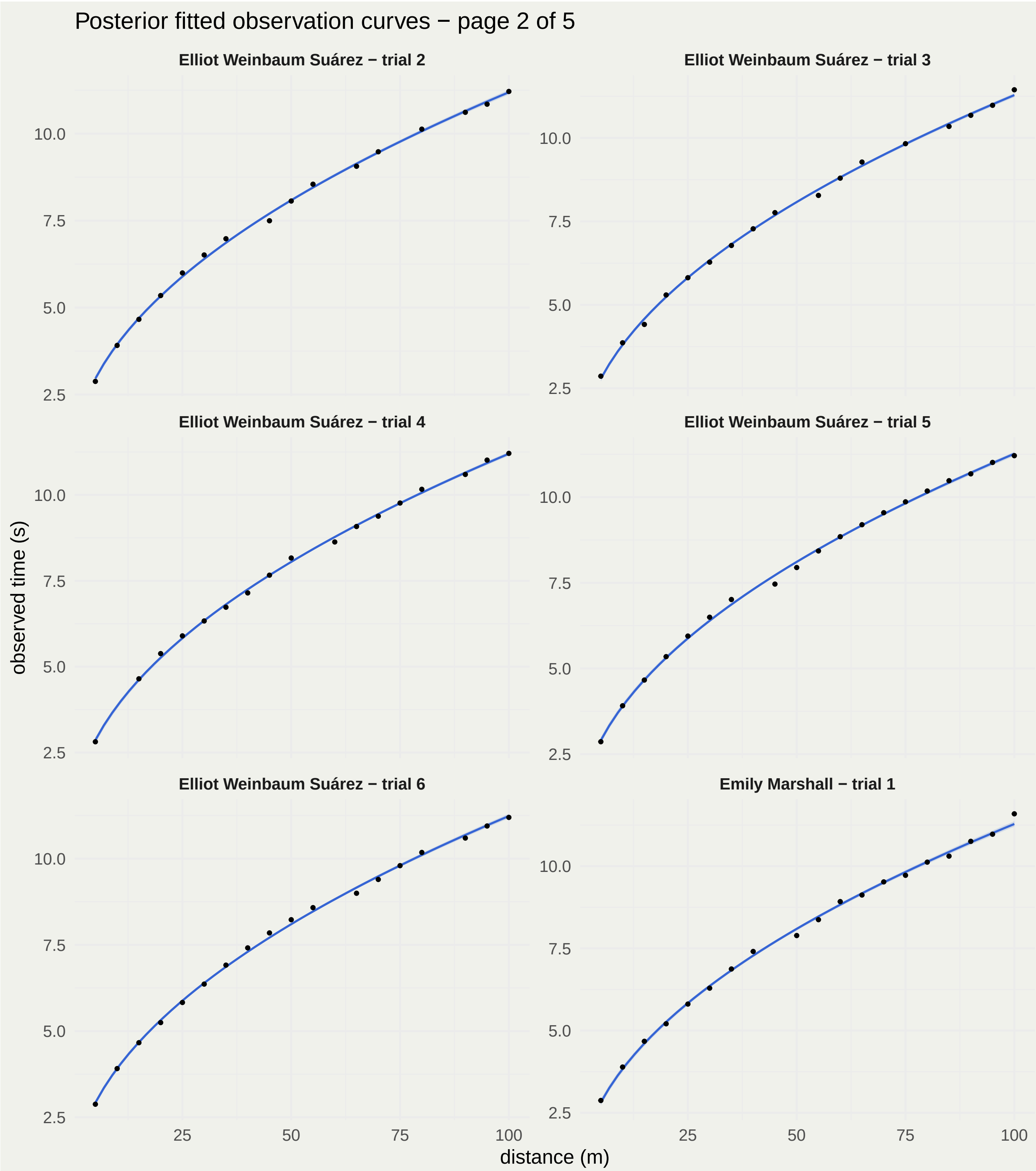
7 Posterior fitted curves for every trial

The following pages show all observations and the posterior fitted observation curve

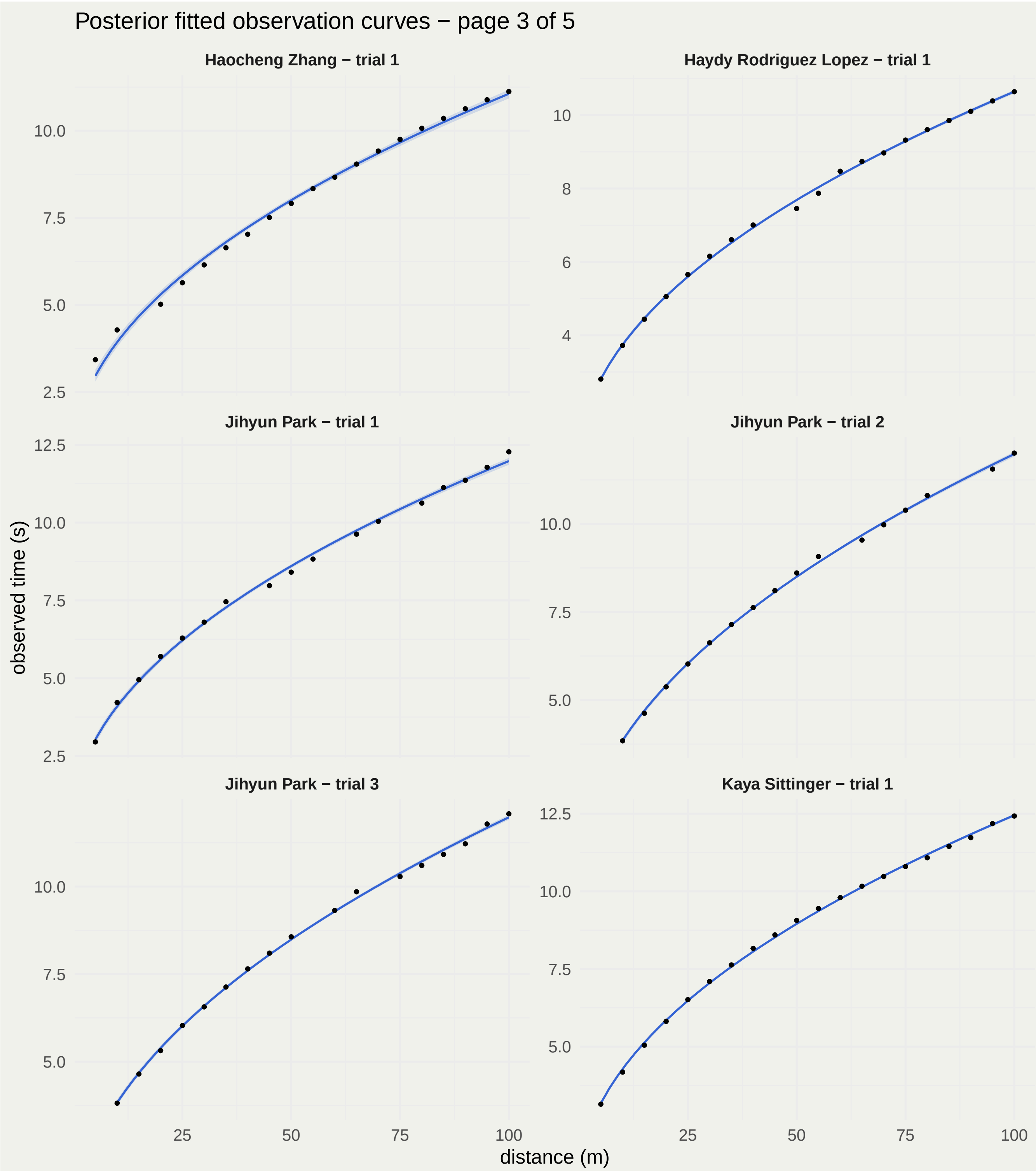
$$\hat{t}^{\text{obs}}(x) = \hat{\alpha} + \hat{\beta}\sqrt{x}$$

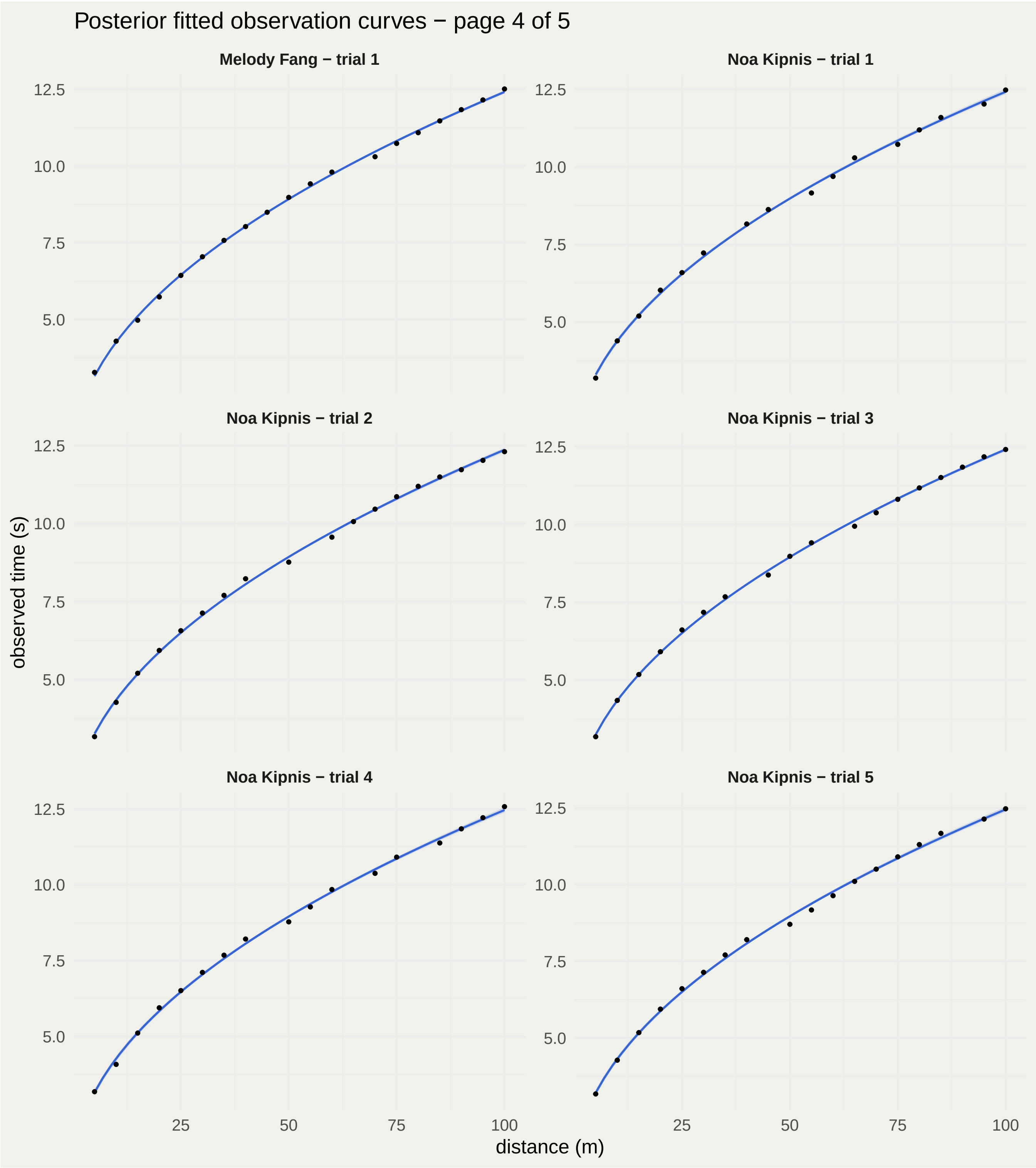
for every student trial. Ribbons are 90% posterior intervals for the fitted mean.

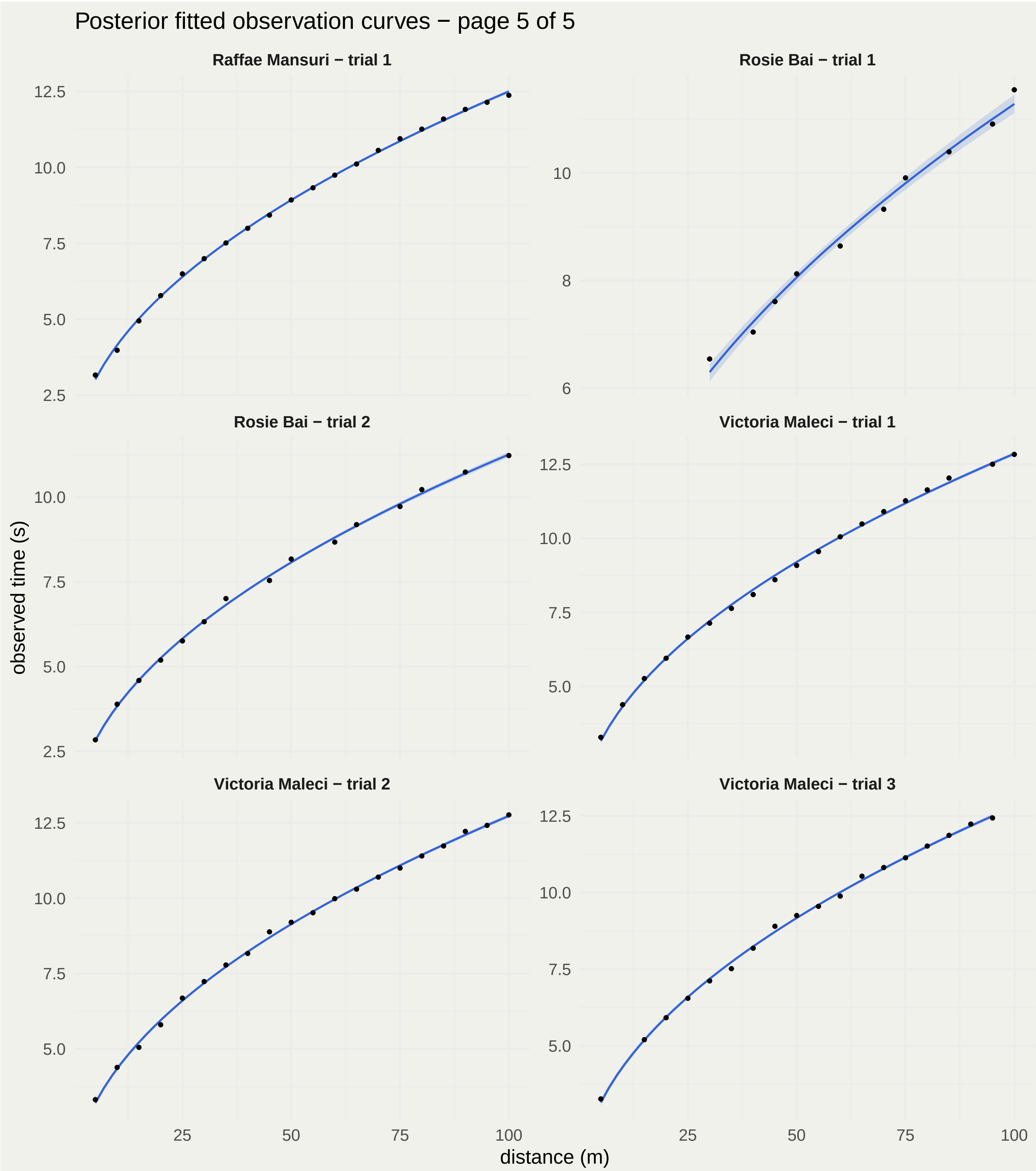

Posterior fitted observation curves − page 1 of 5
Alice Zhang − trial 1
Annie Kiyonaga − trial 1
Annie Kiyonaga − trial 2
Annie Kiyonaga − trial 3
Ben Bozza − trial 1
Elliot Weinbaum Suárez − trial 1
observed time (s)
distance (m)
25
50
75
100



Posterior fitted observation curves − page 2 of 5
Elliot Weinbaum Suárez − trial 2
Elliot Weinbaum Suárez − trial 3
Elliot Weinbaum Suárez − trial 4
Elliot Weinbaum Suárez − trial 5
Elliot Weinbaum Suárez − trial 6
Emily Marshall − trial 1
observed time (s)
distance (m)
2.5
5.0
7.5
10.0
25
50
75
100


Posterior fitted observation curves − page 3 of 5
Haocheng Zhang − trial 1
Haydy Rodriguez Lopez − trial 1
Jihyun Park − trial 1
Jihyun Park − trial 2
Jihyun Park − trial 3
Kaya Sittinger − trial 1
observed time (s)
distance (m)
2.5
5.0
7.5
10.0
12.5
4
6
8
10
25
50
75
100


Posterior fitted observation curves − page 4 of 5
Melody Fang − trial 1
Noa Kipnis − trial 1
Noa Kipnis − trial 2
Noa Kipnis − trial 3
Noa Kipnis − trial 4
Noa Kipnis − trial 5
observed time (s)
distance (m)
12.5
10.0
7.5
5.0
25
50
75
100



8 Conclusions

The labeled instructor runs provide a strong effective-intercept calibration signal:

- Marks runs concentrate near a small intercept.
- Flash runs concentrate near a substantially larger intercept.

- Flash apparent delay decreases during the drop, so the intercept is a mode signature rather than a literal constant delay.

The student ranking is therefore based on the calibrated effective intercept, while the estimated g values and individual plots remain available for interpretation. The trial-level heatmap is especially useful for submissions containing multiple experiments, because it can reveal switches in observation strategy that a student-level average obscures.

9 Appendix A: Rosie Bai source files

The original PNG submission was retained and manually transcribed into a CSV used by the common pipeline. The later submitted CSV records a different experiment and is analyzed as Rosie Bai's trial 2.

TRIAL	MARK	DISTANCE x (m)	OBSERVED TIME t_obs (s)
1	6	30	6.541
1	8	40	7.041
1	9	45	7.608
1	10	50	8.124
1	12	60	8.641
1	14	70	9.324
1	15	75	9.908
1	17	85	10.391
1	19	95	10.908
1	20	100	11.545

Figure 8: Original PNG submission transcribed for analysis.

```r
students |>
  filter(
    student == "Rosie Bai",
    source_format == "PNG transcription"
  ) |>
  select(trial, mark, distance_m, observed_time_s) |>
  knitr::kable(
    booktabs = TRUE,
    caption = "Manual transcription of the PNG submission"
  ) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 9: Manual transcription of the PNG submission

trial	mark	distance_m	observed_time_s
1	6	30	6.541
1	8	40	7.041
1	9	45	7.608
1	10	50	8.124
1	12	60	8.641
1	14	70	9.324
1	15	75	9.908
1	17	85	10.391
1	19	95	10.908
1	20	100	11.545

```
students |>
  filter(
    student == "Rosie Bai",
    source_format == "CSV"
  ) |>
  select(trial, mark, distance_m, observed_time_s) |>
  knitr::kable(
    booktabs = TRUE,
    caption = "Submitted CSV analyzed as Rosie Bai's second experiment"
  ) |>
  kable_styling(latex_options = c("hold_position"))
```

Table 10: Submitted CSV analyzed as Rosie Bai's second experiment

trial	mark	distance_m	observed_time_s
2	1	5	2.8396
2	2	10	3.8906
2	3	15	4.5906
2	4	20	5.1906
2	5	25	5.7573
2	6	30	6.3239
2	7	35	7.0073
2	9	45	7.5406
2	10	50	8.1740
2	12	60	8.6740
2	13	65	9.1900
2	15	75	9.7239
2	16	80	10.2239
2	18	90	10.7406
2	20	100	11.2277

10 Appendix B: Stan source

10.1 Calibration model

```
data {
  int<lower=1> N;
  int<lower=1> J;
  int<lower=1> M;
  array[N] int<lower=1, upper=J> trial;
  array[J] int<lower=1, upper=M> mode;
  vector[N] z;
  vector[N] y;
  vector[J] beta_true;
}

parameters {
  vector[M] mu_alpha;
  real<lower=0> tau_alpha;
  vector[J] alpha_raw;

  vector[M] mu_beta_bias;
  real<lower=0> tau_beta_bias;
  vector[J] beta_bias_raw;

  vector<lower=0>[M] sigma;
}

transformed parameters {
  vector[J] alpha;
  vector[J] beta;

  for (j in 1:J) {
    alpha[j] = mu_alpha[mode[j]] + tau_alpha * alpha_raw[j];
    beta[j] = beta_true[j]
      + mu_beta_bias[mode[j]]
      + tau_beta_bias * beta_bias_raw[j];
  }
}

model {
  mu_alpha ~ normal(0.20, 0.25);
  tau_alpha ~ normal(0, 0.08);
  alpha_raw ~ std_normal();

  mu_beta_bias ~ normal(0, 0.05);
  tau_beta_bias ~ normal(0, 0.04);
  beta_bias_raw ~ std_normal();

  sigma ~ normal(0, 0.15);
```

```
  for (n in 1:N) {
    y[n] ~ normal(
      alpha[trial[n]] + beta[trial[n]] * z[n],
      sigma[mode[trial[n]]]
    );
  }
}

generated quantities {
  vector[J] g_effective;
  vector[N] log_lik;

  for (j in 1:J) {
    g_effective[j] = 2 / square(beta[j]);
  }

  for (n in 1:N) {
    log_lik[n] = normal_lpdf(
      y[n] |
      alpha[trial[n]] + beta[trial[n]] * z[n],
      sigma[mode[trial[n]]]
    );
  }
}
```

10.2 Student-trial model

```
data {
  int<lower=1> N;
  int<lower=1> J;
  array[N] int<lower=1, upper=J> trial;
  vector[N] z;
  vector[N] y;
}

parameters {
  vector[J] alpha;
  vector<lower=0>[J] beta;
  vector<lower=0>[J] sigma;
}

model {
  alpha ~ normal(0.30, 0.50);
  beta ~ normal(1.10, 0.30);
  sigma ~ normal(0, 0.20);

  for (n in 1:N) {
```

```
    y[n] ~ normal(
      alpha[trial[n]] + beta[trial[n]] * z[n],
      sigma[trial[n]]
    );
  }
}

generated quantities {
  vector[J] g_hat;
  vector[N] log_lik;

  for (j in 1:J) {
    g_hat[j] = 2 / square(beta[j]);
  }

  for (n in 1:N) {
    log_lik[n] = normal_lpdf(
      y[n] |
      alpha[trial[n]] + beta[trial[n]] * z[n],
      sigma[trial[n]]
    );
  }
}
```